

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Industry Problem Solving

Code of Conduct:

I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

D. Sindhu.

Signature of the student:

Annexure A

Short Answer Test

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.
2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.
3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.
4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.
5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.
6. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using **K-Nearest Neighbour or Logistic Regression**. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.
7. A smart agriculture system aims to predict crop yield based on soil quality, weather conditions, and irrigation levels. Propose a solution using **Locally Weighted Regression or RBF networks**. Explain how local learning improves predictions and how the system handles seasonal variations and noisy environmental data.
8. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using **KNN or Case-Based Learning**. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.

9. A streaming platform aims to recommend movies to users based on viewing history and ratings. Propose a solution using **KNN or Case-Based Learning**. Explain how user similarity is calculated, how recommendations are generated, and how the system handles large-scale user data efficiently.

10. A transportation company wants to optimize route selection based on traffic, distance, and fuel consumption. Design a solution using **RBF networks or Locally Weighted Regression**. Explain how the model captures complex relationships between variables and improves decision-making in dynamic and real-time environments.

11. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using **K-Nearest Neighbour or Case-Based Learning**. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.

12. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using **Locally Weighted Regression or RBF networks**. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

13. A fraud detection system in an online payment platform needs to classify transactions as legitimate or fraudulent. Design a solution using **KNN or Logistic Regression**. Explain how similarity or decision boundaries are used, how features are selected, and how the model adapts to evolving fraud patterns.

14. A manufacturing company wants to detect defective products based on sensor readings and quality measurements. Propose a solution using **RBF networks or Case-Based Learning**. Explain how non-linear relationships are captured, how similarity between cases is measured, and how the system improves with additional production data.

15. A smart city system aims to predict traffic flow and congestion levels using data from sensors, weather, and time of day. Design a solution using **Locally Weighted Regression or KNN**. Explain how the model captures dynamic patterns, how predictions are made in real-time, and how scalability is maintained with large volumes of data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Answers:

1. Product Recommendation using KNN

Aim:

To recommend products to customers using KNN based on their past purchases.

Approach:

1. Collect customer purchase data.
2. Compare a new customer with existing customers.
3. Find the most similar customers.
4. Recommend products purchased by similar customers.
5. Update the system when new purchases are made.

Analysis:

KNN measures similarity using distance. Similar customers are more likely to prefer similar products.

Advantages: Simple, personalized, easy to update.

Limitation: Can be slow for very large datasets.

Result:

KNN can provide suitable product recommendations based on customer similarity.

2. Machine Output Prediction using LWR

Aim:

To predict machine output using Locally Weighted Regression.

Approach:

1. Collect sensor readings.
2. Find data points close to the new reading.
3. Give higher weight to nearby points.
4. Build a local model.
5. Predict the machine output.

Analysis:

LWR gives more importance to similar and nearby data, so it can handle changing machine behavior.

Advantages: Accurate local prediction and handles changing patterns.

Limitation: Can be slow with large datasets.

Result:

LWR can predict machine output effectively using nearby sensor data.

3. Healthcare Risk Classification using KNN

Aim:

To classify patients into different risk categories using KNN.

Approach:

1. Collect patient medical data.
2. Select important features.
3. Scale the features.
4. Find similar patients using KNN.
5. Assign the patient's risk category.

Analysis:

KNN classifies a new patient based on the risk categories of similar patients.

Advantages: Simple and easy to understand.

Limitation: Performance decreases with very large datasets.

Result:

KNN can classify patients into low, medium, or high-risk categories.

4. Similar Product Detection using Case-Based Learning

Aim:

To identify similar products using Case-Based Learning.

Approach:

1. Store previous product information.
2. Compare a new product with old products.
3. Calculate similarity.
4. Find the most similar products.
5. Use previous pricing information.

Analysis:

The system reuses previous product cases to make decisions for new products.

Advantages: Easy to update and reuses previous knowledge.

Limitation: Large numbers of cases require more searching.

Result:

Case-Based Learning can identify similar products and support dynamic pricing.

5. Delivery Time Prediction using RBF

Aim:

To predict delivery time using an RBF Network.

Approach:

1. Collect distance, traffic, and weather data.
2. Give the data to the RBF network.
3. Learn relationships between the input and delivery time.
4. Predict delivery time for a new order.

Analysis:

RBF networks can learn complex and non-linear relationships between traffic, location, and delivery time.

Advantages: Handles complex patterns and changing conditions.

Limitation: Requires sufficient training data.

Result:

The RBF network can predict delivery time more reliably under different traffic conditions.

6. Loan Default Prediction using KNN / Logistic Regression

Aim:

To identify customers who may default on loans.

Approach:

1. Collect financial and transaction data.
2. Select important features like income, loan amount, and credit history.
3. Use KNN or Logistic Regression.
4. Predict whether the customer will default.
5. Update the model with new customer data.

Analysis:

KNN uses similarity between customers, while Logistic Regression uses a decision boundary to classify customers.

Advantages: Simple and useful for financial prediction.

Limitation: Poor-quality financial data can reduce accuracy.

Result:

The model can identify customers who have a higher risk of loan default.

7. Crop Yield Prediction using LWR / RBF

Aim:

To predict crop yield using soil, weather, and irrigation data.

Approach:

1. Collect soil, rainfall, temperature, and irrigation data.
2. Train an LWR or RBF model.
3. Learn the relationship between environmental factors and crop yield.
4. Predict the yield for new conditions.
5. Update the model with seasonal data.

Analysis:

LWR focuses on similar conditions, while RBF can learn complex relationships.

Advantages: Handles changing environmental conditions.

Limitation: Weather changes can affect prediction accuracy.

Result:

The model can provide useful crop-yield predictions for farmers.

8. Network Intrusion Detection using KNN

Aim:

To detect abnormal network behavior using KNN.

Approach:

1. Collect network traffic data.
2. Extract features such as traffic rate and packet size.
3. Compare new traffic with previous traffic patterns.
4. Detect abnormal patterns.
5. Add new attack data to improve the system.

Analysis:

KNN measures similarity between new network traffic and known patterns. Dissimilar patterns can indicate possible attacks.

Advantages: Simple and can detect different patterns.

Limitation: Can be slow with large network data.

Result:

KNN can identify abnormal network behaviour and help detect intrusions.

9. Movie Recommendation using KNN

Aim:

To recommend movies based on user viewing history and ratings.

Approach:

1. Collect user ratings and viewing history.
2. Compare users based on their preferences.
3. Find similar users.
4. Recommend movies liked by similar users.
5. Update recommendations with new ratings.

Analysis:

KNN calculates similarity between users. Movies watched or highly rated by similar users are recommended.

Advantages: Personalized and simple.

Limitation: New users have little information available.

Result:

KNN can provide personalized movie recommendations based on similar users.

10. Route Selection using RBF / LWR**Aim:**

To select an efficient route using traffic, distance, and fuel information.

Approach:

1. Collect traffic, distance, and fuel data.
2. Train an RBF or LWR model.
3. Predict the cost or travel time of routes.
4. Select the best route.
5. Update the prediction using real-time traffic data.

Analysis:

RBF can learn complex relationships, while LWR focuses on similar traffic conditions.

Advantages: Supports dynamic route selection.

Limitation: Requires continuous traffic information.

Result:

The model helps select efficient routes and reduce travel time and fuel consumption.

11. Data Plan Recommendation using KNN**Aim:**

To recommend suitable mobile data plans based on customer usage.

Approach:

1. Collect customer data usage information.
2. Find customers with similar usage patterns.
3. Identify the plans used by similar customers.
4. Recommend a suitable plan.
5. Update recommendations when usage changes.

Analysis:

KNN compares customers based on data usage, call usage, and other features.

Advantages: Personalized and easy to update.

Limitation: Large customer datasets require efficient searching.

Result:

KNN can recommend suitable data plans according to customer usage patterns.

12. Electricity Consumption Prediction using RBF

Aim:

To predict electricity consumption using time, weather, and appliance usage.

Approach:

1. Collect electricity consumption data.
2. Add weather and time information.
3. Train an RBF network.
4. Predict future electricity consumption.
5. Update the model using new data.

Analysis:

RBF can learn complex relationships between weather, time, appliance usage, and electricity consumption.

Advantages: Handles seasonal and non-linear patterns.

Limitation: Noisy data can affect predictions.

Result:

The RBF model can predict electricity consumption and help improve energy management.

13. Online Payment Fraud Detection using KNN / Logistic Regression

Aim:

To classify online transactions as legitimate or fraudulent.

Approach:

1. Collect transaction data.
2. Select features such as amount, location, and transaction frequency.
3. Train KNN or Logistic Regression.
4. Predict new transactions.
5. Update the model with new fraud cases.

Analysis:

KNN identifies transactions similar to previous fraud cases. Logistic Regression uses a decision boundary to classify transactions.

Advantages: Helps detect fraudulent transactions.

Limitation: Fraud patterns change frequently.

Result:

The system can identify suspicious transactions and improve fraud detection using new data.

14. Defective Product Detection using RBF

Aim:

To detect defective products using sensor and quality measurements.

Approach:

1. Collect sensor readings from production.
2. Select important quality features.
3. Train an RBF network.
4. Classify products as defective or normal.
5. Add new production data to improve the model.

Analysis:

RBF networks can learn non-linear relationships between sensor readings and product quality.

Advantages: Handles complex manufacturing patterns.

Limitation: Requires good-quality training data.

Result:

The RBF model can effectively identify defective products and improve quality control.

15. Traffic Prediction using LWR / KNN**Aim:**

To predict traffic flow and congestion using sensor, weather, and time data.

Approach:

1. Collect traffic sensor data.
2. Include weather and time information.
3. Use LWR or KNN to find traffic patterns.
4. Predict traffic flow in real time.
5. Update predictions using new sensor data.

Analysis:

KNN finds similar previous traffic conditions, while LWR gives more importance to nearby and similar observations.

Advantages: Useful for real-time traffic prediction.

Limitation: Large amounts of traffic data require efficient processing.

Result:

The system can predict traffic flow and congestion, helping smart cities improve traffic management.